Mengkai Hu

Power systems · Vehicle-grid integration · Research tools

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EDUCATION

Tsinghua University | Direct-entry PhD program

Sep 2026 - Present | Supervisor: Prof. Chao Lu

Power Systems and Automation.

Hohai University | Electrical Engineering and Automation

Sep 2022 - Jun 2026 | Supervisor: Prof. Sheng Chen

GPA: 4.99/5.0; ranked 1st in the major. Selected coursework: Power System Analysis, Circuits, Electrical Machines, Engineering Electromagnetics, and Automatic Control.

RESEARCH EXPERIENCE

Distributed photovoltaic and storage planning

National undergraduate innovation project | Project lead | Jun 2023 - Jun 2024

Developed an assessment, integration, and utilization framework for distributed PV. Formulated an MIQCP model with PV and storage operating costs and solved capacity-planning cases on the IEEE 33-bus system using GAMS/GUROBI.

EV user profiling and vehicle-grid interaction

National undergraduate innovation project | First-listed participant | Jun 2024 - Jun 2025

Modeled V2G participation using profiles of ride-hailing vehicles, private cars, and electric buses. Incorporated balking and impatience into an M/M/c/N queuing model and combined charging-load predictions with coordinated V2G optimization.

Grid stability and coordinated control of renewable-energy clusters

China Southern Power Grid research project | Core participant | Jun 2026 - Present

Participates in research on grid-connected stability and coordinated control. A related manuscript on hierarchical neural state-space equations is under review; second author.

GRANTED INVENTION PATENTS

First inventor on both patents listed below.

1. A method for photovoltaic utilization with mobile energy storage based on Dijkstra path planning.

2. A scheduling optimization method for mobile energy-storage vehicles based on charging-station demand and transportation-network equilibrium.

PUBLICATIONS

First author on both publications.

1. Mengkai Hu, Sheng Chen, Sixian Li, Zhinong Wei. Distributionally Robust Coordinated Scheduling of Power-Transportation Systems Considering Bounded Rationality and Flexible Swapping of Electric Vehicles.

IET Energy Systems Integration, 2026, 8(1): e70033. DOI: 10.1049/esi2.70033. CC BY 4.0.

2. Mengkai Hu, Sheng Chen, Zhinong Wei, Si Lyu. Research on collaborative optimization of distribution network considering Vehicle-to-Grid and elastic queuing theory.

Zhejiang Electric Power, 2025, 44(8): 3-14. DOI: 10.19585/j.zjdl.202508001.

MANUSCRIPT UNDER REVIEW

Linear Power System Modeling and Analysis Across Wide Operating Ranges: A Hierarchical Neural State-Space Equation Approach.

IEEE Transactions on Power Systems | Under review | Second author.

OPEN-SOURCE PROJECT

MK Figure

An MIT-licensed toolkit for editable scientific figures, combining a desktop workspace, a Codex skill, and an SVG icon pack, with PPTX, SVG, and PNG export.

SELECTED AWARDS AND HONORS

- National First Prize, 17th National Energy Conservation and Emission Reduction Competition | Aug 2024
- National First Prize, 6th National College Student Renewable Energy Science and Technology Competition | Apr 2024
- Honorable Mention, 40th Mathematical Contest in Modeling | May 2024
- National Scholarship | Academic year 2022-2023
- Baosteel Outstanding Student Award | 2025; Xiaomi Special Scholarship | 2026
- First Prize, Yan Kai Scholarship; Hohai University Outstanding Student Model | 2024

SKILLS

Programming and modeling: C, MATLAB, GAMS, Python.
Engineering software: MATLAB/Simulink, GAMS, Keil5, Visual Studio, Multisim, Altium Designer.
English: CET-6 599; CET-4 603; academic reading and writing.